First-Fit

Input:

user07@jannani:~/firstfit — /usr/bin/vim ff.c

```c
#include<stdio.h>

int main() {
    int m, n;
    printf("Enter number of memory blocks:\n");
    scanf("%d", &m);
    int blocksize[m], remainingsize[m];
    printf("Enter size of each block:\n");
    for (int i = 0; i < m; i++) {
        scanf("%d", &blocksize[i]);
        remainingsize[i] = blocksize[i];
    }
    printf("Enter the number of processes:\n");
    scanf("%d", &n);
    int processsize[n];
    printf("Enter size of each process:\n");
    for (int i = 0; i < n; i++) {
        scanf("%d", &processsize[i]);
    }
    for (int i = 0; i < n; i++) {
        int allocated = 0;
        for (int j = 0; j < m; j++) {
            if (remainingsize[j] >= processsize[i]) {
                int fragment = remainingsize[j] - processsize[i];
                printf("Process %d of size %d is allocated to block %d of size %d with fragment %d\n",
                       i + 1, processsize[i], j + 1, blocksize[j], fragment);
                remainingsize[j] -= processsize[i];
                allocated = 1;
                break;
            }
```

-- INSERT -- 1,18 Top

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```c
        remainingsize[i] = blocksize[i];
    }
    printf("Enter the number of processes:\n");
    scanf("%d", &n);
    int processsize[n];
    printf("Enter size of each process:\n");
    for (int i = 0; i < n; i++) {
        scanf("%d", &processsize[i]);
    }
    for (int i = 0; i < n; i++) {
        int allocated = 0;
        for (int j = 0; j < m; j++) {
            if (remainingsize[j] >= processsize[i]) {
                int fragment = remainingsize[j] - processsize[i];
                printf("Process %d of size %d is allocated to block %d of size %d with fragment %d\n",
                       i + 1, processsize[i], j + 1, blocksize[j], fragment);
                remainingsize[j] -= processsize[i];
                allocated = 1;
                break;
            }
        }


        if (!allocated) {
            printf("Process %d of size %d is not allocated\n", i + 1, processsize[i]);
        }
    }

    return 0;
}
```

-- INSERT -- 40,2 Bo

Output:

```
user07@jannani:~/firstfit
[user07@jannani firstfit]$ gcc ff.c
[user07@jannani firstfit]$ ./a.out
Enter number of memory blocks:
2
Enter size of each block:
150 300
Enter the number of processes:
3
Enter size of each process:
50 70 130
Process 1 of size 50 is allocated to block 1 of size 150 with fragment 100
Process 2 of size 70 is allocated to block 1 of size 150 with fragment 30
Process 3 of size 130 is allocated to block 2 of size 300 with fragment 170
[user07@jannani firstfit]$
```